INCH-POUND

GGG-L-795E
February 21, 1990
(TO SUPERSEDE)
GGG-L-795D
April 25, 1979

FEDERAL SPECIFICATION

LUBRICATING UNIT, POWER OPERATED, AND PUMPING UNITS, PRESSURE LUBRICATING, DRUM MOUNTING

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers lubricating and pumping units, air, electric-motor, and battery powered, for dispensing and transferring general purpose, fibrous, heavy- and light-bodied lubricants from standard open- and closed-top containers for maintenance of trucks and field equipment.

1.1.1 Federal specification coverage. Federal specifications do not include all types, classes, sizes, and styles of lubricating units indicated by the title of the specification, or which are commercially available, but are intended to cover only those types, classes, sizes, and styles which are suitable for Federal Government requirements.

1.2 Classification.

1.2.1 Types, classes, sizes, and styles. The lubricating and pumping units shall be of the following types, classes, sizes, and styles as specified (see 6.2):

Type I - Lubricating unit, air pressure operated (with container cover).
  Class A - High-pressure grease (40 to 1 ratio, minimum).
    Size 1 - 25-pound container.
    Size 2 - 35-pound container.
    Size 4 - 60-pound container.
      Style a - Dolly or chassis mounted.
      Style b - Truck mounted.

Beneficial comments, recommendations, additions, deletions, clarifications, and any data which may improve this document should be sent to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606.

AMSC N/A FSC 4930

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

Size 5 - 120-pound container.
Size 6 - 400-pound container.
Class B - Low-pressure gear oil (2-to-1 to 8-to-1 ratio range).
Size 3 - 40-pound container.
Size 5 - 120-pound container.
Size 6 - 400-pound container.
Class C - High-pressure, high volume.
Size 6 - 400-pound container.

Type II - Lubricating unit, electric-motor-operated (with container cover).
Class A - High-pressure grease.
Size 1 - 25-pound container.
Size 3 - 40-pound container.

Type III - Lubricating unit, battery-powered (with container cover).
Class A - High-pressure grease.
Size 1 - 25-pound container.
Size 2 - 35-pound container.
Size 3 - 40-pound container.

Type IV - Pumping unit, pressure-lubricating, for drum mounting, air operated.
Class A - High-pressure grease (40 to 1 ratio, minimum).
Size 3 - 40-pound container, open top.
Size 5 - 120-pound container, open top.
Size 6 - 400-pound container, open top.
Class B - Low-pressure oil (2-to-1 to 8-to-1 ratio range).
Size 3 - 40-pound container, open top.
Size 5 - 120-pound container, open top.
Size 5 - 120-pound container (with bung opening).
Size 6 - 400-pound container, open top.
Size 6 - 400-pound container (with bung opening).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Federal Specifications

W-C-596	- Connector, Plug, Receptacle and Cable Outlet, Electrical Power, General Specification for.
W-C-596/13	- Connector, Plug, Electrical, General Purpose, Hospital Grade Grounding, 2 Pole, 3 Wire, 15 Amperes, 125 Volts, 50/60 Hertz.
QQ-S-781	- Strapping, Steel, and Seals.
TT-C-490	- Cleaning Methods for Ferrous Surfaces and Pretreatment for Organic Coatings.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:42Z
Check the source to verify that this is the current version before use.

TT-E-489	- Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).
TT-P-664	- Primer Coating, Alkyd, Corrosion-Inhibiting, Lead and Chromate Free, VOC-Compliant.
VV-L-765	- Lubricant, Enclosed-Gear, Nonextreme-Pressure.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed, and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-D-705	- Drum, Shipping and Storage: Steel, 16 and 30 Gallon Capacity.
PPP-D-729	- Drum, Shipping and Storage, Steel, 55 Gallon.
PPP-P-704	- Pails, Metal: (Shipping, Steel, 1 through 12 Gallons).
PPP-T-60	- Tape; Packaging, Waterproof.

Federal Standards

FED-STD-123	- Marking for Shipment (Civil Agencies).
FED-STD-H28	- Screw-Thread Standards for Federal Services.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, and Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Auburn, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and-Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-L-2105	- Lubricating Oil, Gear, Multi Purpose.
MIL-F-3541	- Fittings, Lubrication.
MIL-H-3868	- Hose Assemblies, Non-Metallic, Grease Gun, High and Low Pressure.
MIL-C-4109	- Coupling Halves, Low Pressure, Air Hose, Quick-Disconnect.
MIL-L-4387	- Lubrication Dispensing Equipment Accessories.
MIL-G-10924	- Grease, Automotive and Artillery.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

MIL-H-17672 - Hydraulic Fluid, Petroleum, Inhibited.
MIL-G-81322 - Grease, Aircraft, General Purpose, Wide Temperature Range, NATO Code Number G-395.

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-838 - Lubrication of Military Equipment.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474 - Noise Limits for Army Materiel.
MS15000 - Fittings, Lubrication-Hydraulic, Acceptable Tip Designs.

Military Handbook

MIL-HDBK-113 - Guide for the Selection of Lubricants, Functional Fluids, Preservatives and Speciality Products for Use in Ground Equipment Systems.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

Federal Regulations

Code of Federal Regulations (CFR)

48 CFR 23.4 - Use of Recovered Materials.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:42Z
Check the source to verify that this is the current version before use.

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA 390.03 - Gear Handbook, Volume 1.
AGMA 201.02 - Tooth Proportions for Coarse-Pitch Involute Spur Gears.

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Avenue, NW, Washington, DC 20005.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1 - Structural Welding Code, Steel.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33126.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG-1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

SOUTH COAST AIR QUALITY MANAGEMENT CONTROL DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management Control District, 9150 East Flair Drive, El Monte, CA 91731.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

3. REQUIREMENTS

3.1 Description. Lubricating and pumping units (hereinafter called "units"), air, electric, and battery power operated (with exceptions, as specified), shall consist of a chassis mounted on two wheels or casters, with one swivel-type caster, a handle, lubricant container, pressure pump, discharge hose, control valve, lubricating accessories, safety valves, racks, and power supply cord.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

Figures are intended for guidance and identification (see 6.9). Specific requirements shall be as specified herein.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.3 Design. Units shall be designed to satisfactorily dispense chassis grease conforming to MIL-G-10924, arctic oils conforming to MIL-L-2105, or general purpose aircraft grease conforming to MIL-G-81322, as applicable. Components shall be of such size, material, and strength as to properly sustain the maximum allowable load imposed upon them with an adequate factor of safety, maximum working efficiency, and minimum wear. Units and their parts shall be complete so that when connected to the power supply line, they can be used for any operation for which designed.

3.3.1 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.3.2 Threads. All threads shall conform to FED-STD-H28.

3.3.3 Fasteners. All screws, bolts, and similar parts shall be installed with a means for preventing loss of tightness and adjustment. All such parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently fastened. All fasteners shall be wet assembled.

3.3.4 Gears. All gears and pinions shall be in accordance with AGMA 390.03. Spur gear tooth form shall conform to AGMA 201.02.

3.3.5 Lubrication. All bearings and surfaces requiring lubrication shall be provided with a means for lubricating.

3.3.5.1 Lubricants. The procedure for the selection of lubricants shall conform to MIL-STD-838, section 5. Lubricants selected shall be as specified in MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a Qualified Products List (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.3.6 Packing and gaskets. Packing and gaskets shall be of corrosion- and grease-resistant material capable of withstanding all the pressures and operating conditions specified herein.

3.3.7 Electrical connections. Electrical connections shall be mechanically secured and soldered or brazed, except connections on screw-type terminals, which shall be mechanically secured.

3.4 Materials. Materials shall be as specified herein. Materials shall be free from all defects and imperfections that might affect the serviceability or safety characteristics of the finished product. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.4.1 Material deterioration prevention and control. The units shall be fabricated from compatible materials, inherently corrosion resistant or treated

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the units may be exposed.

3.4.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.1.2 Identification of materials and finishes. The contractor shall identify the specific materials, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4.2 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 (48 CFR 23.4) to the maximum extent practical.

3.5 Finish. Nonworking surfaces subject to atmospheric corrosion shall be given a pretreatment, primer, and enamel finish conforming to TT-C-490, TT-P-664, and TT-E-489, respectively.

3.6 Environmental conditions.

3.6.1 Storage temperatures. Each unit shall be capable of withstanding storage without damage in a temperature of -60 °F to 150 °F.

3.6.2 Operating temperature. Each unit shall be capable of operating effectively when subjected to an operating temperature of -20 °F to 120 °F.

3.7 Safety. Each unit shall be effectively guarded and grounded, as required, to protect all persons and objects from electrical shock hazard. All belts, pulleys, and other reciprocating, rotating, or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard.

3.8 Electromagnetic interference. When specified (see 6.2), electromagnetic interference emission characteristics of the units shall conform to MIL-STD-461, class IIB.

3.9 Accessories. When specified (see 6.2), high-pressure metering guns, hydraulic couplers, and swivel connectors shall be finished as specified in MIL-L-4387.

3.10 Fungus- and moisture-resistance. When specified (see 6.2), the electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.10.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

a. Components or circuit elements that are inherently fungus- and moisture-resistant or which are hermetically sealed need not be treated.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:42Z
Check the source to verify that this is the current version before use.

b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter that would interfere with the adherence or function of the varnish.

3.10.1 Composition II. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material conforms to Rule 442, South Coast Air Quality Management Control District.

3.11 Containers. Unless otherwise specified (see 6.2 and 6.4), a container shall be furnished with each type I, II, and III unit. Containers up to and including 60-pound capacity shall conform to 3.14.1.6.2.5 or 3.14.1.6.2.6, as applicable. Containers of 120-pound capacity shall conform to PPP-D-705, type III. Containers of 400-pound capacity shall conform to PPP-D-729, as applicable. (Note: Containers are not furnished with type IV units.)

3.12 Storage and discharge. When specified (see 6.2), the type I, class A, sizes 5 and 6, and the type I, class B, sizes 5 and 6 units shall meet the storage and discharge tests of 4.5.2.2 and the delivery requirements of table I.

TABLE I. Delivery requirements.

Requirement	Type I, class A, sizes 5 and 6	Type I, class B, sizes 5 and 6
Rate of discharge to atmosphere at 25 °F ±2 °F at 150 psi with 20 feet of hose.	14 ounces per minute of grease conforming to MIL-G-10924.	1.5 quarts per minute of SAE grade 140 lubricant.
Rate of discharge to atmosphere at 115 °F ±2 °F at 150 psi with 20 feet of hose.	16 ounces per minute of grease conforming to MIL-G-10924.	4 quarts per minute of SAE grade 140 lubricant.

3.13 Performance. Performance characteristics shall be as specified in table II.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:42Z
Check the source to verify that this is the current version before use.

TABLE II. Performance.

Requirement	Type I, class A	Type I, class B	Type I, class C	Type II, class A	Type III, class A,	Type IV, class A	Type IV, class B
Minimum bursting pressure of hose (see note)	12,000 psi	3,000 psi	12,000 psi	12,000 psi	8,000 psi 3/	12,000 psi	3,000 psi
Pressure on hose connections without evidence of leakage	6,000 psi	1,500 psi	6,000 psi	6,000 psi	6,000 psi	6,000 psi	1,500 psi
Rate of discharge to atmosphere at 70 °F ±2 °F through grease outlet of pump (minimum)	14 oz. per min. using air pressure of 100 psi 1/	8.5 lbs per min. using air pressure of 100 psi 2/	6 lbs per min. using air pressure of 100 psi 1/	9 oz. per min. when operating at normal voltage frequency 1/	40 lbs of lubricant per hour 1/	25 oz. per min. using air pressure of 100 psi 1/	7 lbs. per min. using air pressure of 100 psi 2/
Rate of discharge (at 20 °F) w/100 psi air pressure and w/3000 psi back pressure	8 oz. per minute	-	28 oz. per minute	5 oz. per minute	-	8 oz. per minute	-
Static pressure developed by pump without evidence of leakage (minimum)	6,000 psi with air pressure of 150 psi	1,200 psi with air pressure of 150 psi	6,000 psi with air pressure of 150 psi	6,000 psi operating at normal voltage and frequency	6,000 psi	6,000 psi with air pressure of 150 psi	1,200 psi with air pressure of 150 psi

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

1/ Grease conforming to MIL-G-10924.
2/ SAE 90 motor oil.
3/ In the event the booster pressure of 12,000 psi passes through the hose, the minimum bursting pressure of the hose shall be 12,000 psi.
(Note: See MIL-H-3868 for military hose requirements.)

3.14 Type I, air-pressure operated.

3.14.1 Class A, high pressure. Type I, class A, high-pressure air-operated units shall be suitable for maintenance of trucks and all-around lubrication in repair shops. They shall be capable of handling fibrous, heavy- and light-bodied lubricants. Each unit shall include an air-operated pump as specified in 3.14.1.1. The hose assembly and mounting shall be as specified in 3.14.1.3. Containers shall conform to 3.11. Operating characteristics shall be as specified in table II.

3.14.1.1 Pump. Each unit shall be provided with an air-operated high-pressure pump of the single cylinder type capable of delivering lubricant on both the up and down strokes. The pump shall have a minimum ratio of 40 to 1 and shall be capable of producing a discharge pressure of not less than 6,000 psi at the control valve when operating on an air-pressure of 150 psi. The air motor shall be equipped with self-adjusting "O" rings or packing. The foot of the pump shall be fitted with a 30-mesh wire screen or other filtering device to prevent entrance of foreign matter into the pump. When a screen is furnished, the screen shall be made of corrosion-resisting wire or wire with a corrosion-resistant plating. The screen or filtering device shall be readily removable, and it shall not be necessary to disassemble any part of the pump for cleaning purposes. The pump shall be of the plunger type with a built-in priming device to ensure delivery of grease under conditions specified in table I or II, as applicable. Pumps shall be constructed of materials of such strength and proportion to withstand the pressure developed, with a suitable factor of safety. The outlet shall have either a 1/4-, 3/8-, or 1/2-inch internal pipe thread. A nozzle holder shall be provided and suitably mounted for attaching the hose assembly swivel. Pumps shall be capable of pumping lubricants to within a maximum of 2 inches from the bottom of containers.

3.14.1.2 Air motor. The motor shall be of the compressed air-driven type suitable for operation on air pressure of 80 to 150 psi and shall withstand stall pressure up to 200 psi without damage. It shall operate without excessive vibration at any load within the rated capacity of the unit. A quick-disconnect type coupler shall be provided at the air inlet of the motor. Motors requiring daily lubrication with oil shall be clearly and prominently marked "OIL DAILY". Motors which require lubricating with grease shall be clearly and prominently marked. The exterior working parts of the motor shall be protected from dust and dirt.

3.14.1.2.1 Air consumption. Compressed air consumption at 70 °F, with 100 psi air-line pressure shall not exceed 6 cubic feet of compressed air per minute when pumping the lubricant for which the pump was designed at free flow through hose and control valve.

3.14.1.3 High-pressure lubricant discharge hose. Unless a 10- or 20-foot minimum hose length is specified (see 6.2), each unit shall be equipped with a 6-foot minimum length of high-pressure lubricant hose. The hose shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

measured from the lubricator coupling to the lubricant control valve and shall have either a nominal 1/4-, 3/8-, or 1/2-inch inside diameter. The hose shall conform to MIL-H-3868, type I, grade B, or standard commercial hose which meets the test requirements of MIL-H-3868 (see 6.2). A swivel fitting conforming to MIL-L-4387, type VII, class 2, size as specified (see 6.2), shall be provided to connect to the pump outlet and to connect various attachments to the opposite end.

3.14.1.4 Control valve. The control valve shall conform to MIL-L-4387, type VI, class 1, style A, except the inlet shall have internal threads to mate with the swivel fitting. When specified (see 6.2), a suitable pressure-reducing device shall be provided in or with the aforementioned control valve of such nature that when the device is set in the "low-pressure position", the pump pressure ratio will be reduced to approximately 10 to 1 or 8 to 1 ratio or 8 times the air pressure supplied to operate the pump, and when the device is set in the "high-pressure position", the full pump pressure (40 to 1 ratio, minimum) shall be available.

3.14.1.5 Hydraulic coupler. The coupler shall be of a type designed for effecting a leak-free connection between the lubricator and hydraulic-type fittings conforming to MIL-F-3541. The coupler shall maintain a tight seal, without external assistance, when coupled with the specified fittings, and subjected to an internal lubricant pressure, and shall maintain such seal when the centerline of the coupler is displaced at any angle up to and including 10 degrees in any direction from the vertical centerline of the lubrication fitting. The coupler shall be so designed as to grip the fitting with a pressure in proportion to the pressure of the lubricant being discharged, and shall operate satisfactorily at any pressure for which the unit is designed. It shall require a straight pull of approximately 80 pounds, or an angular displacement of the coupler with reference to the vertical centerline of the fitting between 15 and 35 degrees to separate the coupler from the fitting. Connection between the coupler and the fitting shall be affected by a straight push over the fitting of not more than 20 pounds. The jaws shall be reversible and a minimum of 0.050-inch in thickness. The coupler jaws and nozzle shall be case hardened to a minimum depth of 0.003-inch. The coupler sleeve shall be made of a hardened alloy steel.

3.14.1.6 Container sizes and mounting for type I, class A units.

3.14.1.6.1 Size 1, 25-pound container and size 2, 35-pound container. The pump and a 25-pound or 35-pound container with lid (see figure 1) shall be mounted on a suitable base. The base shall be fitted with either 3 or 4 casters, or 2 wheels with either 1 or 2 front casters which shall permit easy movement over rough floors. When specified (see 6.2), a handle may be provided for carrying in lieu of pulling the container. A rigid pulling handle shall be provided when the base is equipped with casters or wheels. The container shall not leak when tested as specified in 4.5.2.11.

3.14.1.6.2 Size 4, 60-pound container (nominal).

3.14.1.6.2.1 Pump mounting. The pump shall be mounted on and form a part of the cover. It shall be of the single cylinder type capable of delivering lubricant on both the up and down strokes. It shall be so designed that it will exhaust grease from the container to within not more than 2 inches of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

bottom. The pump shall be designed for a ratio of not less than 40 to 1, that is, it shall deliver at a pressure of not less than 40 times the air pressure used and shall be suitable for satisfactory operation on air from 80 to 150 psi. The air motor shall be equipped with self-adjusting "O" rings or packing.

3.14.1.6.2.2 Dimensions. The unit and its mounting (dolly, chassis, or truck) shall have overall dimensions of not more than 44 inches high by 26 inches by 26 inches.

3.14.1.6.2.3 Racks. The unit shall be equipped with racks or other means for coiling of the dispensing hose. A nozzle recess or other suitable means shall be provided on the unit for holding the nozzle when it is not in use.

3.14.1.6.2.4 Cover. The cover shall be made of sheet metal either 18 gauge (0.0478-inch) minimum if not reinforced, or 22 gauge (0.0299-inch) minimum if reinforced.

3.14.1.6.2.5 Style A, dolly- or chassis-mounted. Style A units and parts thereof, shall be mounted on a dolly (see figure 2) or chassis (see figure 3). Parts shall be protected from leakage or spillage of lubricant during use. The dolly or chassis shall be mounted on three or more wheels or casters. Each wheel or caster shall be not less than 2.5 inches in diameter and shall have a minimum load capacity of 70 pounds and shall be designed to swivel and roll freely on suitable bearings and shall be equipped with rubber tires (synthetic or natural), plastic tires, or steel tires, as specified (see 6.2). Style A units shall be of the type suitable in all respects for using refinery packages of 25-, 30-, 35-, and 50-pound capacity. The cover shall be so designed that it may be securely attached to a dolly or chassis, or drum if the drum is securely attached to the dolly or chassis, but should be readily removed when so desired.

3.14.1.6.2.6 Style B, truck-mounted. Style B units (see figure 4) and parts thereof, shall be mounted on a two-wheel truck. Parts shall be protected from leakage or spillage during use. The wheels shall be not less than 8 inches in diameter and shall have a minimum load carrying capacity of 75 pounds each and shall be equipped with rubber tires. A suitable handle shall be provided on the truck for convenience in moving from place to place. The lubricating container shall be of the type suitable for holding 25-, 35-, and 50-pound capacity refinery packages or not less than 60 pounds of bulk lubricant. The cover shall be fastened in place on the container by a quick-fastening clamp or other equally suitable means. A rubber sealing ring or similar material shall be used to make an air- and dust-tight joint between the cover and container at atmospheric pressures. The cover shall be so designed that it may be easily and quickly removed from the container.

3.14.1.7 Size 5, 120-pound container. The pump mounting (see figure 1) shall be of a design suitable for use as a cover for a regular 120-pound drum. There shall also be furnished a dolly or base with three or more casters, of adequate size and construction to permit easy movement over rough floors.

3.14.1.8 Size 6, 400-pound container. The size 6 container (see figure 5) shall be for stationary use. The pump shall be suitable for mounting in the bunghole of a standard 400-pound drum. When specified (see 6.2), a drum cover for open-head 400-pound drums shall be furnished.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:42Z
Check the source to verify that this is the current version before use.

3.14.2 Class B, low pressure. Type I, class B, low-pressure air-operated units shall be suitable for pumping semifluid greases into gear cases, etc. Each unit shall consist of an air-operated pump, air motor, hose assembly, and mounting, as specified in 3.14.2.1 through 3.14.2.4. The units shall be capable of delivering lubricant at the rate and under the conditions specified in tables I and II, as applicable.

3.14.2.1 Pump. The pump shall be an air-operated plunger type with a foot valve to ensure proper priming. The grease outlet shall have either a 3/8- or 1/2-inch internal pipe thread.

3.14.2.2 Air motor. The air motor shall be as specified in 3.14.1.2.

3.14.2.3 Hose assembly. One hose assembly complete with control valve, meter, and swivel as specified in 3.14.2.3.1 through 3.14.2.3.3 shall be furnished with each size 3 unit.

3.14.2.3.1 Hose. The hose shall be not less than 5 feet in length, 1/2-inch inside diameter, low pressure, grease pump discharge type. When specified (see 6.2), a 10- or 20-foot length of hose shall be furnished in lieu of the 5-foot hose. The interior of the hose shall be of suitable synthetic material, and the exterior of oil-resistant material. Fittings shall be of the swaged-on type and shall not blow off or form blisters at the connections between hose and fittings when subjected to maximum pressure. Fittings at each end shall have pipe threads.

3.14.2.3.2 Control valve and meter. The control valve and meter shall consist of a hand valve for controlling the flow of lubricant and a dial or digital readout to register and totalize the amount delivered. The lubricant outlet shall be fitted with a nondrip nozzle. Unless otherwise specified (see 6.2), the nozzle shall be either of the double-swivel type or provided with a flexible hose in the meter. At the option of the contractor, the totalizing meter may be mounted on the pumping unit.

3.14.2:3.3 Swivel. The swivel shall be low-pressure-volume type. It shall permit easy handling of the control valve and prevent strains on the hose.

3.14.2.4 Mounting. The mounting shall conform to 3.14.1.7 for size 5.

3.14.3 Class C, high pressure, high volume, size 6. Class C high-pressure, high-volume, air-operated units shall be suitable for maintenance of trucks and general lubrication in repair shops. These units shall be designed for mounting on a standard 400-pound drum with center opening. They shall be capable of handling fibrous, heavy- and light-body lubricants and shall meet the operating characteristics specified in table II. Each unit shall consist of a pump, air motor, and pump mounting as specified in 3.14.3.1 through 3.14.3.3, and, unless otherwise specified (see 3.11), a container.

3.14.3.1 Pump. The pump shall be an air-operated, plunger-type capable of delivering the applicable grade of grease under the conditions specified in table II. The grease outlet shall have either a 3/8- or 1/2-inch internal pipe thread. The pump shall be similar to figure 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

3.14.3.2 Air motor. The air motor shall be as specified in 3.14.1.2.

3.14.3.3 Pump mounting. The mounting shall be of a design suitable for use as a cover for a regular 400-pound capacity drum with center opening.

3.14.3.4 Hose assembly. The hose assembly shall include 20 feet of high pressure hose, and control valve with hydraulic coupling. The hose shall be of the grease-pump, discharge-type; interior of the hose shall be suitable synthetic material, and the exterior of oil-resistant material. Fittings shall be of the swaged-on type and shall not blow off or form blisters at the connections between hose and fittings when subjected to maximum pressure. For military acquisition, hose and fittings shall conform to MIL-H-3868, type I, grade B (see 6.2). Hoses furnished to the Government shall be not more than 12 months old from date of cure of the rubber to date of acceptance by the Government.

3.15 Type II, electric-motor operated.

3.15.1 Class A, size 1. Electric-motor operated, high-pressure units (see figure 6) shall be suitable for maintenance of trucks and all-around lubrication in repair shops. They shall consist of a motor-operated pump, hose assembly, and 25-pound container with lid, mounted on a suitable base, as specified in 3.15.1.1 through 3.15.1.5, inclusive.

3.15.1.1 Electrical.

3.15.1.1.1 Motor. Unless otherwise specified (see 6.2), the motor shall conform to NEMA MG-1, alternating current (ac), 120 volts, not less than 0.375 horsepower (hp), 60 Hertz (Hz), single-phase, continuous duty, and explosion proof.

3.15.1.1.2 Electrical components. Electrical components of units, including relays, switches, wiring, and their enclosures, shall conform to, and be located as specified in the applicable NEMA electrical standards, except as specified herein. No component or part shall be used in any application where the requirements exceed the manufacturer's rating for the part. Electrical components shall be explosion proof.

3.15.1.1.3 Power supply cord. A 3-conductor, type S, heavy-duty power supply cord not less than 25 feet long as specified in NFPA No. 70 shall be furnished with each unit. The free end of the cord shall be provided with a connector plug conforming to W-C-596/13.

3.15.1.1.4 Dielectric strength. The electrical circuits of the unit shall be capable of withstanding, without damage, the application of 1,000 volts root mean square (rms), 60 Hz, ac, plus twice the normal operating voltage for not less than 1 minute, when applied between the frame and current-carrying parts.

3.15.1.2 Pump. The pump shall handle all types of light-bodied lubricants. It shall be fitted with a grease strainer and check valve.

3.15.1.3 Hose assembly. The hose assembly shall include 6 feet minimum of high-pressure hose, and control valve with hydraulic coupling, except, when specified (see 6.2), a 10-foot minimum length of hose shall be furnished. The hose shall be of the grease-pump, discharge-type; interior of the hose shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

of suitable synthetic material and the exterior of oil-resistant material. Fittings shall be of the swaged-on type and shall not blow off or form blisters at the connections between hose and fittings when subjected to maximum pressure. For military acquisition, hose and fittings shall conform to MIL-H-3868, type I, grade B.

3.15.1.4 Container. The container shall not leak and shall incorporate a device to wipe the lubricant from the sides of the container and to force the lubricant from the container to the pressure pump.

3.15.1.5 Base. The base on which the pump and container are mounted shall be of metal and fitted with a handle and either three casters or two wheels and one front caster to permit easy movement over rough floors.

3.15.2 Class A, size 3. Class A, size 3 electric-motor-operated lubricating units shall conform to class A, size 1, units except the container shall have a 40-pound capacity.

3.16 Type III, battery powered.

3.16.1 Class A, high-pressure. Type III, class A, battery powered, high-pressure units (see figure 7) shall be suitable for lubrication of equipment in the field. Each unit shall consist essentially of a pump, motor, pressure-control device, lubricant container cover, hose assembly, battery cable assembly and, unless otherwise indicated by the option in 3.11, a grease container with handle. When specified (see 6.2), a battery compatible with the electrical requirements of the unit shall also be furnished.

3.16.1.1 Pump. The pump shall be of a self-priming type and shall not air-lock under normal operating conditions. A relief valve shall be provided for releasing any air-lock in the event the pump develops such air-lock under adverse operating conditions. A fine screen filter shall be provided to filter the lubricant before it enters the pumping mechanism. Means shall be provided whereby the filter can be easily cleaned. The pump shall operate efficiently in the container sizes specified in table III.

TABLE III. Container size.

Size	Container capacity in pounds (minimum)	Container depth in inches (approx.)
1	25	9.5
2	35	11
3	40	13

3.16.1.2 Motor. Unless otherwise specified (see 6.2), the motor shall be suitable for operation on 12-volt batteries ordinarily used in automotive and field equipment. The motor shall operate the gun for 4 hours of continuous duty without discharging below 1.200 specific gravity. When specified (see 6.2), the motor shall be provided with an insulating ground lead to prevent discharging of the battery and other damage in the event of improper installation of the unit.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:42Z
Check the source to verify that this is the current version before use.

3.16.1.3 Pressure control device. A device shall be furnished consisting of a solenoid, oil-pressure gauge, and a pressure switch, or any other suitable means whereby an easy adjustment can be made to obtain any desired line pressure from a low pressure of 400 psi or less up to a high pressure of 3,000 psi or more. All components shall withstand the rated line pressure specified in table II. If any higher line pressure than that specified in table II is furnished, the components shall withstand the rated pressure without evidence of leakage.

3.16.1.4 Container cover. The container cover shall be of all metal construction and provided with adjusting screws or other means whereby the cover can be fitted to a commercial grease container.

3.16.1.5 Hose assembly. The hose assembly shall consist essentially of a hose and control valve mechanism.

3.16.1.5.1 Hose. The hose shall be of a flexible synthetic material that is impervious to oils and greases. The hose shall be not less than 25 feet in length and shall withstand the pressure specified in table III. For military acquisition, the hose and fittings shall conform to MIL-H-3868, type I, grade B (see 6.2).

3.16.1.5.2 Control valve mechanism. The control valve mechanism shall provide for swiveling adjustment and easy, one-hand operation. The control valve mechanism shall include a component that is adaptable to standard grease fittings on automotive chassis and farm equipment. The component shall be suitable for assembly of special fittings (see 6.3). The control valve shall be equipped with a mechanical booster capable of increasing the line pressure from 3,000 psi to 12,000 psi.

3.16.1.6 Battery cable assembly. The battery cable assembly shall be suitable for use with the voltage requirements of the motor and battery and shall consist of a negative and a positive lead, 10-feet minimum length, with easily attached terminal connections.

3.17 Type IV, pumping unit.

3.17.1 Type IV, pumps, lubricant. The units of this type shall be reciprocating, self-priming, air motor-driven, composed primarily of a motor section and pump section. Compressed air consumption, at 100 psi line pressure, shall not exceed 4 cubic feet per minute when pumping the lubricant for which the pump is designed at free lubricant flow, in quantities as hereinafter specified for each class and style, at ambient room temperature of 70 °F. Connection between the compressed air line and the air motor shall be accomplished through an air line quick-disconnect coupler. For military purchases, the coupler shall conform to MIL-C-4109, type I, 1/4-inch size. Unless otherwise specified (see 6.2), each pump motor shall be furnished with at least 5 feet of 1/4- or 5/16-inch inside diameter, 600 psi minimum burst pressure air hose with 1/4-inch hose external NPTF end fittings. Pumps shall be of the single- or double-acting types, as hereinafter specified for each class. Single-acting pumps shall discharge material being pumped on one of the strokes of the complete cycle. Double-acting pumps shall discharge material on both strokes of the complete cycle. The pressure ratio of all pumps covered by this specification shall be considered as the ratio between the air pressure at the motor inlet and the discharge pressure of the pump when gauged at stall pressure. All pumps shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

capable of pumping all the lubricant from the drums to within a maximum depth of 2 inches from the bottom. Type IV pumps for bung mounting shall be similar to figure 5, and type IV pumps for open top mounting shall be similar to figure 5, except that the drum cover is not shown in the figure.

3.17.2 Type IV, class A, pumps, lubricant, high-pressure grease. Pumps of this type and class shall be of the double-acting type with a minimum pressure ratio of 40 to 1 for pumping general purpose greases, and grease conforming to MIL-G-10924, from full, open drums conforming to PPP-P-704, PPP-D-705, and PPP-D-729, as applicable. Unless otherwise specified (see 6.2), the pump shall be furnished complete with drum cover and follower plate to fit the aforementioned drums; also, with at least 5 feet of either 1/4-inch or 3/8-inch inside diameter lubricant hose, provided with either 1/4-inch or 3/8-inch external NPTF end fittings conforming to MIL-H-3868, type I, style 2, grade B, for use in effecting connection between the pump and pipeline delivery service. Pumps of this type and class shall deliver a minimum of 25 ounces per minute of grease conforming to MIL-G-10924, when operated under conditions specified in 3.16.1.

3.17.3 Type IV, class B pumps, lubricant, low-pressure oil. Pumps of this type and class shall be of the single- or double-acting type with a pressure ratio of between 2 to 1 and 8 to 1 for pumping gear lubricants conforming to VV-L-765 and oil conforming to MIL-H-17672 through the bung or open top in the drum head of containers conforming to PPP-P-704, PPP-D-705, and PPP-D-729, as applicable. Unless otherwise specified (see 6.2), the pump shall be furnished complete with bung adaptor to fit the drum or a drum cover, as applicable, and at least 5 feet of 1/2- or 7/16-inch inside diameter hose with 3/8-inch or 1/2-inch external NPTF end fittings conforming to MIL-H-3868, type II, style 1, grade B for use in effecting connection between the pump and pump line delivery service. Pumps of this type and class shall deliver a minimum of 7 pounds of grade 90 gear lubricant, as specified in VV-L-765, and 1-1/2 gallons of number 2135 oil, as specified in MIL-H-17672, per minute when operated under conditions specified in 3.17.1.

3.18 Human factors engineering. The lubricating units shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to general requirements (4), visual displays (5.2), controls (5.4), labeling (5.5), design for maintainability (5.9), hazards and safety (5.13) of MIL-STD-1472, as applicable.

3.19 Noise limits. The noise produced by the lubricating units shall conform to MIL-STD-1474, category E requirements, with the exception of MIL-STD-1474, 5.2, 5.3, and 5.4, when tested in accordance with 4.5.2.13.

3.20 Identification marking. Units shall be permanently marked with the manufacturer's name or a trademark of such known character that the source of manufacturer and lot may be readily identified and shall include the specification part number (see 6.6).

3.21 Workmanship. Units shall conform to the quality and grade of product established by this specification. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and subassemblies; thoroughness of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

soldering, brazing, welding, riveting, and painting; alignment of parts and tightness of screws, bolts, and other fastenings.

3.21.1 Castings and forgings. All parts, components, and assemblies, which include castings and forgings, shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.21.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the units to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.1875 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the units shall be at the option of the contractor. Components which are to be welded which have an operating pressure equal to or more than 15 psig shall be welded in accordance with ASME Boiler and Pressure Vessel Code, Section IX. Components other than these shall be with either AWS D1.1 or the ASME welding code.

3.21.3 Welders. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Structural Welding Code, AWS D1.1, Steel.

ASME Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". In the event of poor welds, the Government reserves the right to require retesting of any welder or welding operator. All test results, welding procedures and procedure qualification records shall be made available to the contracting officer or his representatives upon request.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined, and tested as specified in referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of preparation for delivery (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article shall be tested as specified in 4.5.2.1 through 4.5.2.13. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall conform to MIL-STD-105.

4.4.2 Examination. Samples selected as specified in 4.4.1 shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. Samples selected as specified in 4.4.1 shall be tested as specified in 4.5.2.1, 4.5.2.3, 4.5.2.4, 4.5.2.6, 4.5.2.7 (omit 4.5.2.7.1), and 4.5.2.8 through 4.5.2.11. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. Units shall be examined as specified herein for the following defects:

101. Design not as specified (see 3.3).
102. Parts interchangeability not as specified (see 3.3.1).
103. Threads not as specified (see 3.3.2).
104. Fasteners not as specified (see 3.3.3).
105. Gears not as specified (see 3.3.4).
106. Lubrication not as specified (see 3.3.5).
107. Lubricants not as specified (see 3.3.5.1).

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:42Z
Check the source to verify that this is the current version before use.

108. Packing and gaskets not as specified (see 3.3.6).
109. Electrical connections not as specified (see 3.3.7).
110. Materials not as specified (see 3.4).
111. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.4.1).
112. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.4.1.1).
113. Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.4.1.2).
114. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the units (see 3.4.2).
115. Finish not as specified (see 3.5).
116. Safety features not as specified (see 3.7).
117. When required, electromagnetic interference characteristics not as specified (see 3.8).
118. When required, accessories missing or not as specified (see 3.9).
119. When required, fungus and moisture resistance not as specified (see 3.10, 3.10.1).
120. When required, container missing or not as specified (see 3.11).
121. Components missing or not as specified (see 3.14 through 3.17.3).
122. Functional components require abnormal force to operate (see 3.14 through 3.17.3).
123. When required, electrical motors and components not explosion proof (see 3.15.1.1.1, 3.15.1.1.2).
124. When required, power supply cord not as specified (see 3.15.1.1.3).
125. When required, battery missing or not as specified (see 3.16.1).
126. Motor not as specified (see 3.16.1.2).
127. Battery cable assembly missing or not as specified (see 3.16.1.6).
128. Human factors engineering not as specified (see 3.18).
129. Identification marking missing, incomplete, or illegible (see 3.20).
130. Workmanship not as specified (see 3.21).
131. Castings and forgings not as specified (see 3.21.1).
132. Welding not as specified (see 3.21.2).
133. Welder or welding operator qualifications not as specified (see 3.21.3).

4.5.2 Tests.

4.5.2.1 Operation tests. Unless otherwise specified (see 6.2), tests shall be made at atmospheric pressure (approximately 29.92 inches Hg.) and at room temperature (approximately 70 °F). When tests are made with atmospheric pressure or room temperature differing materially from the specified values, proper allowance shall be made for the difference from the specified condition. The performance checks of the displaced fluid shall be determined by comparison of actual weight of the displaced fluid versus time.

4.5.2.2 Storage and discharge tests.

4.5.2.2.1 Low-temperature test. The unit shall be placed in an ambient air temperature of -60 °F to -65 °F for a period of 72 hours. At the end of this time, the unit shall be brought to 25 °F and tested for the delivery requirement of table I. Nonconformance to table I shall constitute failure of this test (see 3.6.1, 3.6.2 and table I).

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:42Z
Check the source to verify that this is the currentversion before use.

4.5.2.2.2 High-temperature test. The unit shall be submitted to a 24-hour cycle of temperature starting with 10 hours at 90 °F, followed by 5 hours of steady increase to 150 °F, 4 hours at 150 °F, and 5 hours of steady decrease to 115 °F. The unit shall then be tested for the delivery requirement of table I. Nonconformance to table I shall constitute failure of this test (see 3.6.1, 3.6.2 and table I).

4.5.2.2.3 Discharge capacity test. The unit with lubricant shall be stabilized at the test temperature. During this test, the lubricant shall be discharged into a container of known weight and volume. The hydraulic coupler shall be attached to a hydraulic fitting and the unit operated free flow for a 5-minute period. The lubricant dispensed shall be weighed to determine conformance to 3.11. Nonconformance to 3.11 shall constitute failure of this test.

4.5.2.3 Control-valve pressure. The coupler at the end of the control valve shall be attached to a lubricating fitting mounted on a pressure gauge with not less than 0 to 15,000 psi dial reading. The unit shall be operated for a continuous pumping period of not less than 5 minutes. The pressure gauge shall indicate a pressure of not less than that shown in table II for the type unit being tested. Indication of a pressure less than that shown in table II shall constitute failure of this test.

4.5.2.4 Bursting pressure test. The hose shall be subjected to internal pressure, for the particular type and class, as specified in table II. Any evidence of bubbles, breaks, or leaks shall constitute failure of this test.

4.5.2.5 Bursting pressure and hose connection tests for military purchases. Extra lengths of hose and fittings, as required, shall be supplied for destructive testing to conform to MIL-H-3868. Extra hose lengths shall be sufficient to replace all burst hose. The Government inspector may permit reduced hose testing as appropriate when small unit lots are submitted for testing (see 3.14.1.3, 3.14.2.3.1, 3.14.3.4, 3.15.1.3 and 3.16.1.5.1).

4.5.2.6 Hose-connection test. The hose and connections shall be subjected to internal pressures, for the particular type and class, as specified in table II. Any evidence of leakage shall constitute failure of this test.

4.5.2.7 Rate of discharge. A clean receptacle shall be weighed and the weight recorded. The hydraulic coupler shall be attached to a lubricating fitting. For military use, the fitting shall conform to MS15000. The receptacle shall then be placed at the grease outlet. The unit shall be operated in cycles of 3 minutes "ON" and 1 minute "OFF" for a total of 10 cycles. Grease discharged on the first, fifth, and ninth cycles shall then be weighed to determine conformance to the requirements of table II. Noncompliance with the requirements of table II shall constitute failure of this test.

4.5.2.7.1 Delivery against pressure test. Each type and class of unit specified in table I requiring 200 psi and 3,000 psi back pressure shall be filled with the lubricant specified in table II and then maintained at a temperature of 20 °F ±2 °F for a period of 12 hours. The unit shall be operated with an air pressure of 100 psi while maintaining the specified back pressure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

The discharged lubricant shall be collected in a tare container and weighed. A discharge of less than the amounts specified in table II shall constitute failure of this test.

4.5.2.8 Static-pressure test for grease-dispensing units. When the unit is supplied with hose assembly, the hydraulic coupler from the control valve shall be engaged with a standard lubricating fitting mounted on a pressure gauge having suitable dial reading. With the control valve open, the pump shall be operated at the static lubricant pressure specified in the applicable part of table II. There shall be no leakage from the pump, hose assembly fittings or around the hydraulic coupler. For testing the type I, class A, size 4, or type I, class C, size 3 pump, the pressure gauge shall be mounted directly in the grease outlet and the unit operated as specified in the applicable part of table II without evidence of leakage. Any evidence of leakage shall constitute failure of this test.

4.5.2.8.1 Air leakage test. When the type I, class A or C unit is operated as specified in 4.5.2.8, it shall be observed for air leakage from the air motor. Evidence of leakage so as to affect operation shall constitute failure of this test (see table II).

4.5.2.9 Leakage test for oil-dispensing units. The type I, class B gun supplied with a hose assembly shall be mounted on an oil container ready for use. The pump shall be run under the conditions specified in table II for determining the rate of discharge. The control valve shall be fully opened and closed five times using quick pulls and releases on the valve trigger to produce maximum pressure fluctuations in the oil supply hose. The control valve shall be held open long enough to permit full flow of the lubricant. There shall be no leakage in any of the hose assembly fittings around the pump or around the air motor. The type I, class B pump supplied without a hose assembly shall be tested by closing the pump outlet with a 1/2-inch pipe plug. The pump shall be operated under the conditions specified above without evidence of leakage around the pump or air motor. Any evidence of leakage shall constitute failure of this test (see table II).

4.5.2.10 Meter test for oil-dispensing units. The accuracy of the meter and totalizer supplied with the type I, class B units shall be determined by operating the pump under the conditions specified in 4.5.2.9. The lubricant shall be discharged into a graduated container. The column indicated on the meter and totalizer shall not differ from the measured volume delivered by more than 0.25 pint for delivered quantities up to 1 gallon. Evidence to the contrary shall constitute failure of this test. The meter shall hold its accuracy when the control valve is opened abruptly. Evidence that the meter does not hold its accuracy when the control valve is opened abruptly shall constitute failure of this test (see 3.14.2.3.2).

4.5.2.11 Container tightness. Each container shall be filled with free-flowing lubricant and examined for evidence of leakage. Evidence of leakage shall constitute failure of this test (see 3.14.1.6, 3.15.1.4).

4.5.2.12 Electromagnetic interference. When required (see 6.2), the unit shall be tested as specified in MIL-STD-462 to determine conformance to 3.8. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

4.5.2.13 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments, one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of preparation for delivery.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall conform to MIL-STD-105.

4.6.1.3 Examination. Samples selected as specified in 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

134. Parts removed that will require special tools for reinstallation.
135. Unprotected surfaces not coated with preservative as specified.
136. Openings into motors, pumps, containers, and hoses not covered or wrapped with barrier material as specified.
137. Accessories not preserved and consolidated as specified.
138. Technical publications not preserved as specified.
139. Strapping not as specified.
140. Marking illegible, incorrect, incomplete, or missing.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. When disassembly is considered necessary for reduction of cubage, the disassembly shall be limited to parts that can be removed and reinstalled without special tools.

5.1.1.2 Cleaning. Units shall be cleaned in accordance with MIL-P-116, Process C-1.

5.1.1.3 Drying. Units shall be dried in accordance with MIL-P-116.

5.1.1.4 Unprotected surfaces. Working surfaces of the air motor, pump, control valve, and swivels requiring a contact preservative as specified in MIL-P-116, shall be coated with type P-2 preservative. Exterior surfaces of the units requiring a contact preservative as specified in MIL-P-116 shall be coated with type P-1 preservative.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

5.1.1.5 Openings. All openings of the air and electric motors, pumps, containers, and hoses shall be covered or wrapped with barrier material conforming to MIL-B-121, type II, grade A, class 2. The barrier material shall be secured with tape conforming to PPP-T-60, type IV.

5.1.1.6 Accessories. When furnished, accessories shall be preserved as specified in MIL-P-116. The preserved accessories shall be consolidated in a close-fitting box conforming to PPP-B-636, class weather resistant. Cushioning and blocking shall be provided in accordance with MIL-STD-1186 to prevent or control movement within the box.

5.1.1.7 Technical publications. Technical publications shall be preserved as specified in MIL-P-116, method IC-1.

5.1.2 Level C. Units and accessories shall be preserved as specified for level A except the boxes shall be domestic class.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Each complete unit, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional. Blocking and cushioning shall be provided in accordance with MIL-STD-1186 to prevent movement within the container. Strapping shall conform to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for level A except that boxes shall conform to PPP-B-601, domestic type or PPP-B-621, class 1, and strapping may be finish A.

5.2.3 Level C. Units preserved as specified in 5.1 shall be packed in wooden boxes that will prevent damage to the contents during shipment. Cushion, anchor, block, and brace in accordance with MIL-STD-1186.

5.3 Marking.

5.3.1 Civil agencies. Marking shall conform to FED-STD-123.

5.3.2 Military packaging. Marking shall conform to MIL-STD-129.

6. NOTES

6.1 Intended use. The lubricating and pumping units covered by this specification are intended for dispensing and transferring general purpose, fibrous, heavy- and light-bodied lubricants from standard open- and closed-top containers for maintenance of trucks and field equipment.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:42Z
Check the source to verify that this is the current version before use.

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Specification part number of the unit required (see 6.6 and 1.2).

d. When a first article is not required (see 3.2).

e. Timeframe required for submission of first article model and number of units required (see 3.2).

f. When electromagnetic interference emission characteristics are required (see 3.8).

g. When and which accessories are required (see 3.9).

h. When fungus- and moisture-resistance are required (see 3.10).

i. When container for type I, II, or III units will not be required (see 3.11 and 6.4).

j. When table I delivery requirements will be required (see 3.12).

k. When a 10-foot or 20-foot minimum length of high-pressure hose will be required for types I or II, class A units (see 3.14.1.3 and 3.15.1.3).

l. Size of swivel fitting required (see 3.14.1.3).

m. Color of hose, if other than as specified (see 3.14.1.3, 3.14.2.3.1, 3.14.3.4, 3.15.1.3, 3.16.1.5.1).

n. When a pressure-reducing device will be required on type I, class A units (see 3.14.1.4).

o. When a carrying handle will be acceptable for type I units with a 25-pound container (see 3.14.1.6.1).

p. When rubber tires, plastic tires, or steel tires will be required for type I, class A units (see 3.14.1.6.2.5).

q. When drum cover for open-head, 400-pound drum will be required (see 3.14.1.8).

r. When a 10-foot or 20-foot length of hose will be required for type I, class B units, state length required (see 3.14.2.3.1).

s. When a specific nozzle assembly will be required on type I, class B units, state whether "double-swivel type" or "flexible-hose type" will be required (see 3.14.2.3.2).

t. Motor, if other than specified (see 3.15.1.1.1).

u. When compatible battery will be required (see 3.16.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

v. When operating voltage other than 12-volts will be required (see 3.16.1.2).

w. When an insulated ground lead will be required on motors for type III units (see 3.16.1.2).

x. Length and types of hose for type IV units, if different (see 3.17.1).

y. When hose and drum accessories will not be required for type IV pumps (see 3.17.2 and 3.17.3).

z. When welding process will not be at option of contractor (see 3.21.2).

aa. Operational test, if different (see 4.5.2.1).

bb. When electromagnetic interference testing will be required (see 4.5.2.12).

cc. Level of preservation and packing required (see 5.1 and 5.2).

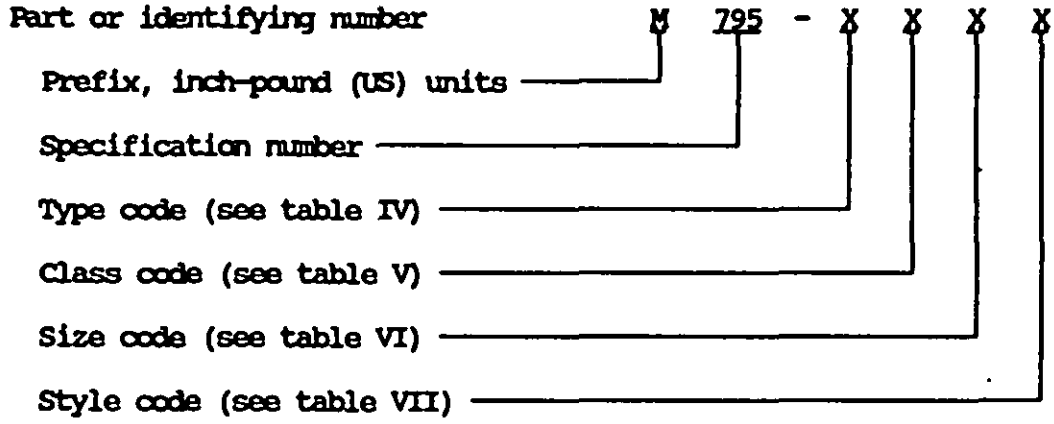
6.3 Special fittings for type III guns. When special fittings are required for type III guns (see 3.16.1.5.2), the procuring agency should furnish the contractor with all necessary data for procurement such as the trade name, size, etc., of the grease fittings to which the special fittings are to be adapted.

6.4 Container option. The description for lubricating units (types I, II, and III herein) stipulates a storage tank, whereas the description for pumping units (type IV herein) stipulates a pumping unit designed to clamp on lubricant drums. In commercial practice, some lubricating units are not normally furnished with drums; therefore, the procuring agency should use care in exercising the option in 3.11. The reason some lubricating units are not furnished with containers is that the unit mounts directly onto regular commercial lubricant containers.

6.5 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Part or identifying number (PIN). The definitive specification part number for the lubricating and pumping units covered by this specification shall be designated in the following form (see 1.2 and 6.2).

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:42Z
Check the source to verify that this is the current version before use.



Example: M795-IA4b is the definitive specification part number for a truck mounted (b), 60-pound container (4), high-pressure grease (A), air pressure operated lubricating unit (I) covered by this specification (795) measured in inch-pound units (M).

6.6.1 Type code. See table IV.

TABLE IV. Type code.

Code (Type)	Description
I	Lubricating unit, air pressure operated (with container cover).
II	Lubricating unit, electric-motor-operated (with container cover).
III	Lubricating unit, battery-powered (with container cover).
IV	Pumping unit, pressure-lubricating, for drum mounting, air operated.

6.6.2 Class code. See table V.

TABLE V. Class code.

Code (Class)	Description
A	High-pressure grease (40 to 1 ratio, minimum).
B	Low-pressure gear oil (2-to-1 to 8-to-1 ratio range).
C	High-pressure, high volume.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

6.6.3 Size code. See table VI.

TABLE VI. Size code.

Code (Size)	Description
1	25 pound container.
2	35 pound container.
3	40 pound container.
4	60 pound container.
5	120 pound container.
6	400 pound container.

6.6.4 Style code. See table VII.

TABLE VII. Style code.

Code (Style)	Description
a	Dolly or chassis mounted.
b	Truck mounted.
c	Open top.
d	With bung opening.

6.7 Technical manuals. When this specification is cited in a contract, a contract exhibit must be prepared to fully describe statement of work criteria and delivery instructions and cite this specification. Technical manuals must be acquired by separate contract line item number (CLIN) in the contract. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each unit.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of units.

6.9 Information figures. Figures show types of units which have been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other units which conform to this specification.

6.10 Lubricants. MIL-STD-838 Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113 Guide for the Selection of Lubricants, Functional Fluids, Preservatives and Speciality Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development, and Engineering Center

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

(ATTN: STRBE-VF), Fort Belvoir, VA 22060-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2 (see 3.3.5.1).

6.11 Subject term (key word) listing.

Lubricating units, air powered
Lubricating units, battery powered
Lubricating units, electric-motor powered
Pumping units, air powered
Pumping units, battery powered
Pumping units, electric-motor powered

6.12 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence.

MILITARY INTERESTS:

Custodians
Army - ME
Navy - SH
Air Force - 99

Review Activity
DLA - CS

User Activities
Navy - MC, SH

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Army - ME

Project 4930-0319

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:42Z
Check the source to verify that this is the current version before use.

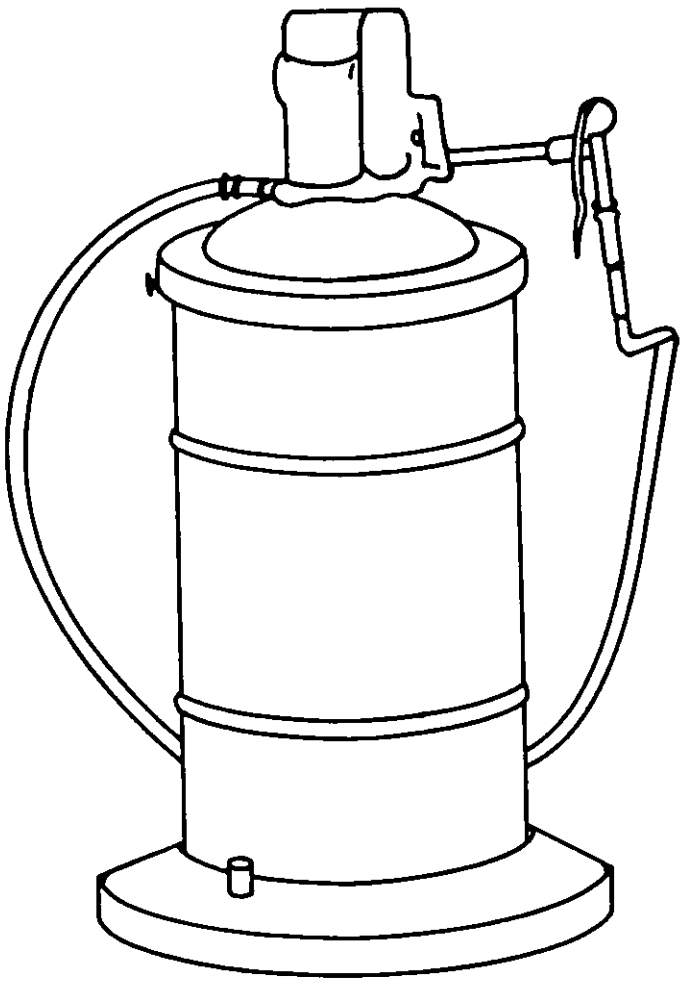


FIGURE 1. Type I, class A, sizes 1, 2, and 5 and class B, sizes 3 and 5.

X-3366

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:42Z
Check the source to verify that this is the current version before use.

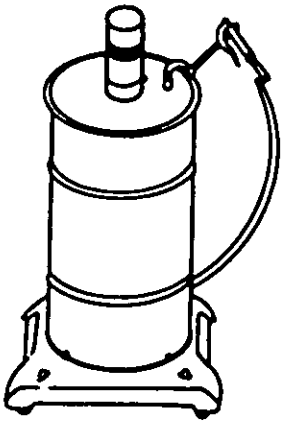


FIGURE 2. Dolly-mounted, type I, class A, size 4, style a.

X-3367

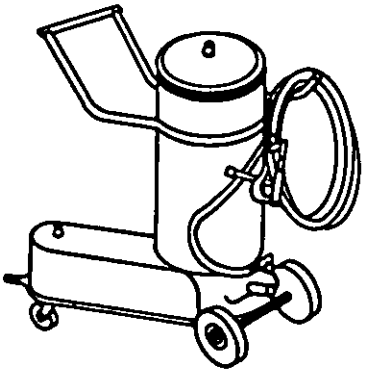


FIGURE 3. Chassis-mounted, type I, class A, size 4, style a.

X-3368

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:42Z
Check the source to verify that this is the current version before use.

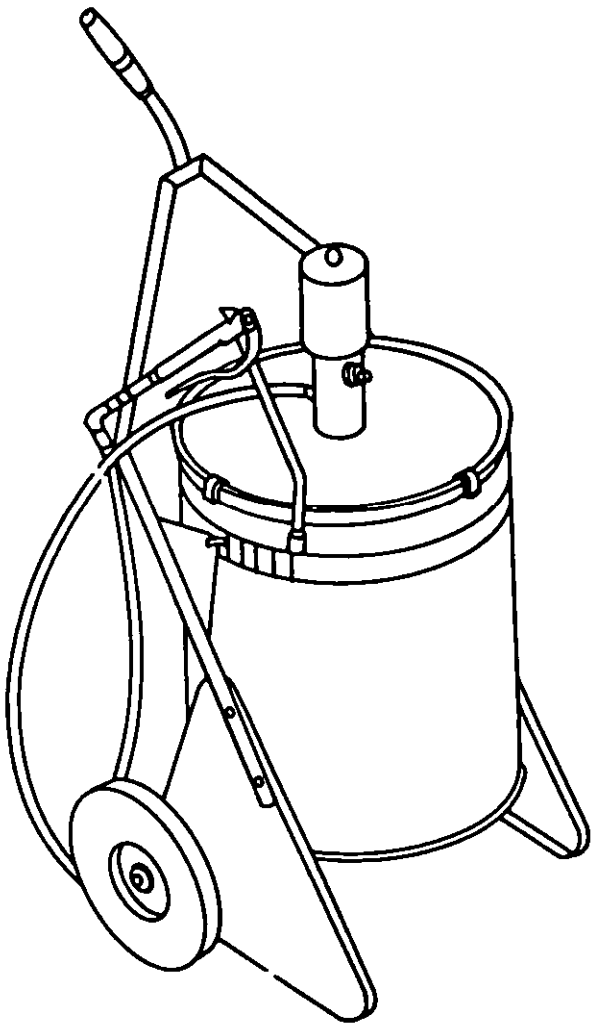


FIGURE 4. Truck-mounted, type I, class A, size 4, style b.

X-3369

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

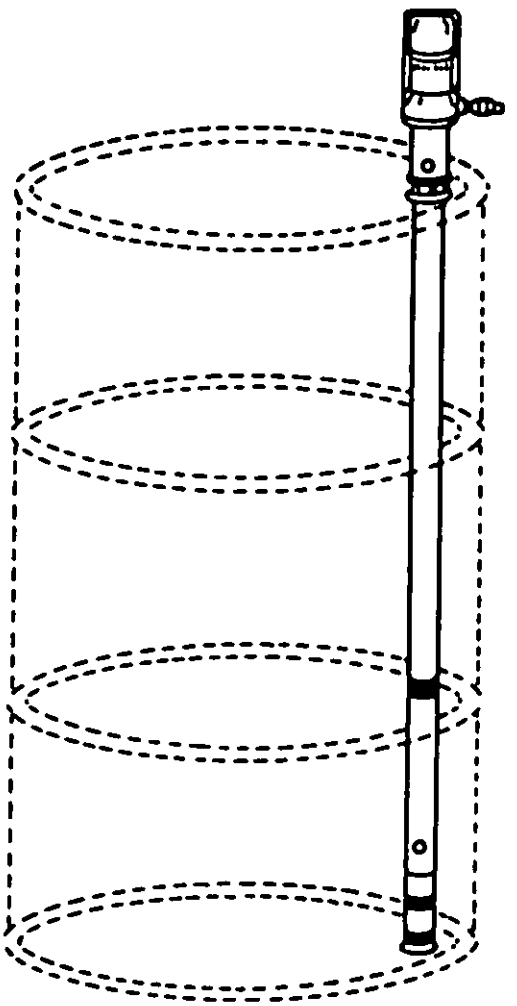


FIGURE 5. Type I, class A, size 6 and classes B and C, size 6.

X-3370

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

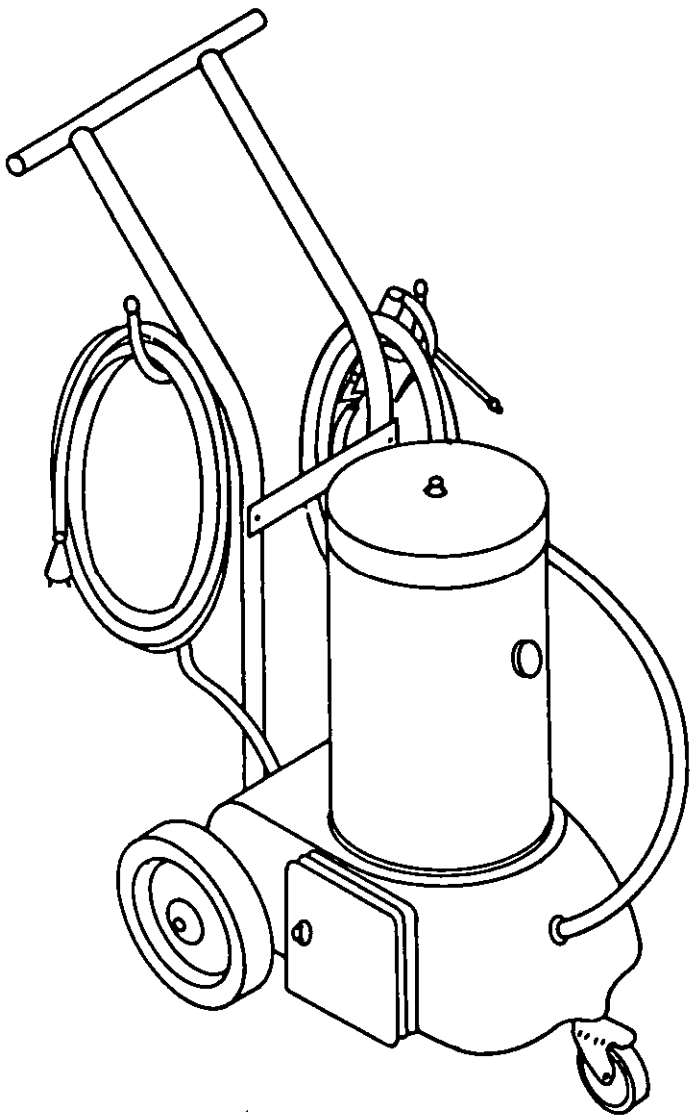


FIGURE 6. Type II, class A, sizes 1 and 3, electric-motor operated.

X-3371

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

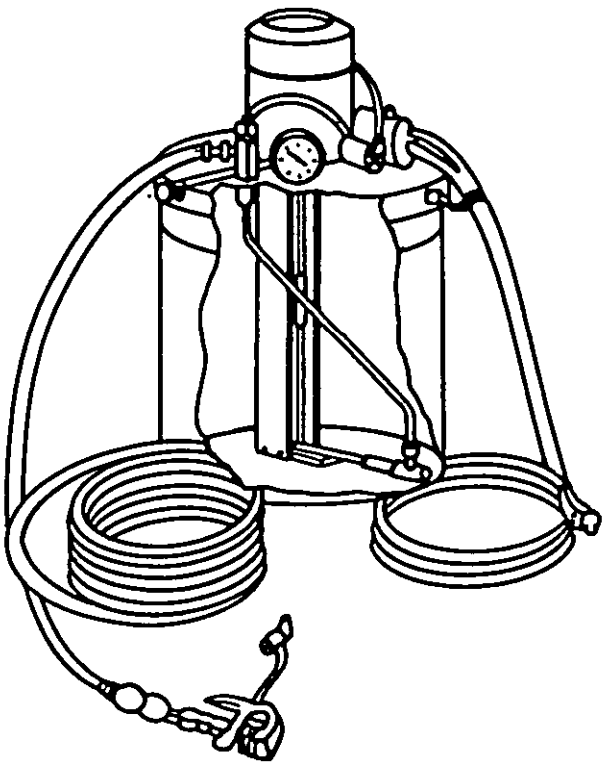


FIGURE 7. Type III, class A, sizes 1, 2, and 3, battery powered.

X-3372

☆U.S. GOVERNMENT PRINTING OFFICE: 1990 - 784-634/20040

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER GGG-L-795E	2. DOCUMENT DATE (YYMMDD) 21 Feb 90

3. DOCUMENT TITLE: Lubricating Unit, Power Operated, and Pumping Units, Pressure Lubricating, Drum Mounting

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (*Include Area Code*) (1) Commercial: (703) 664-5717 (2) AUTOVON: 354-5717
c. ADDRESS (*Include Zip Code*) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:42Z
Check the source to verify that this is the current version before use.